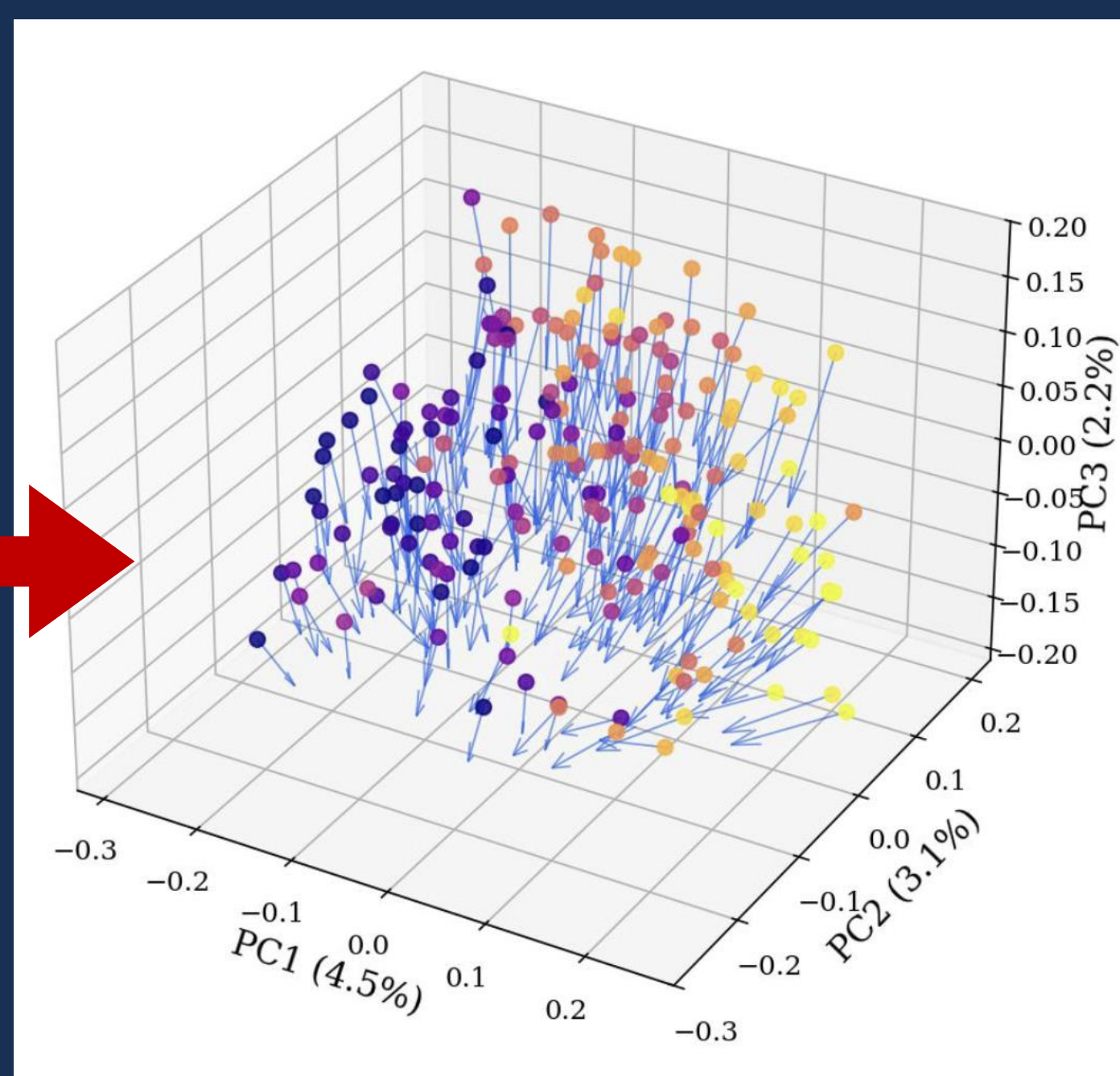
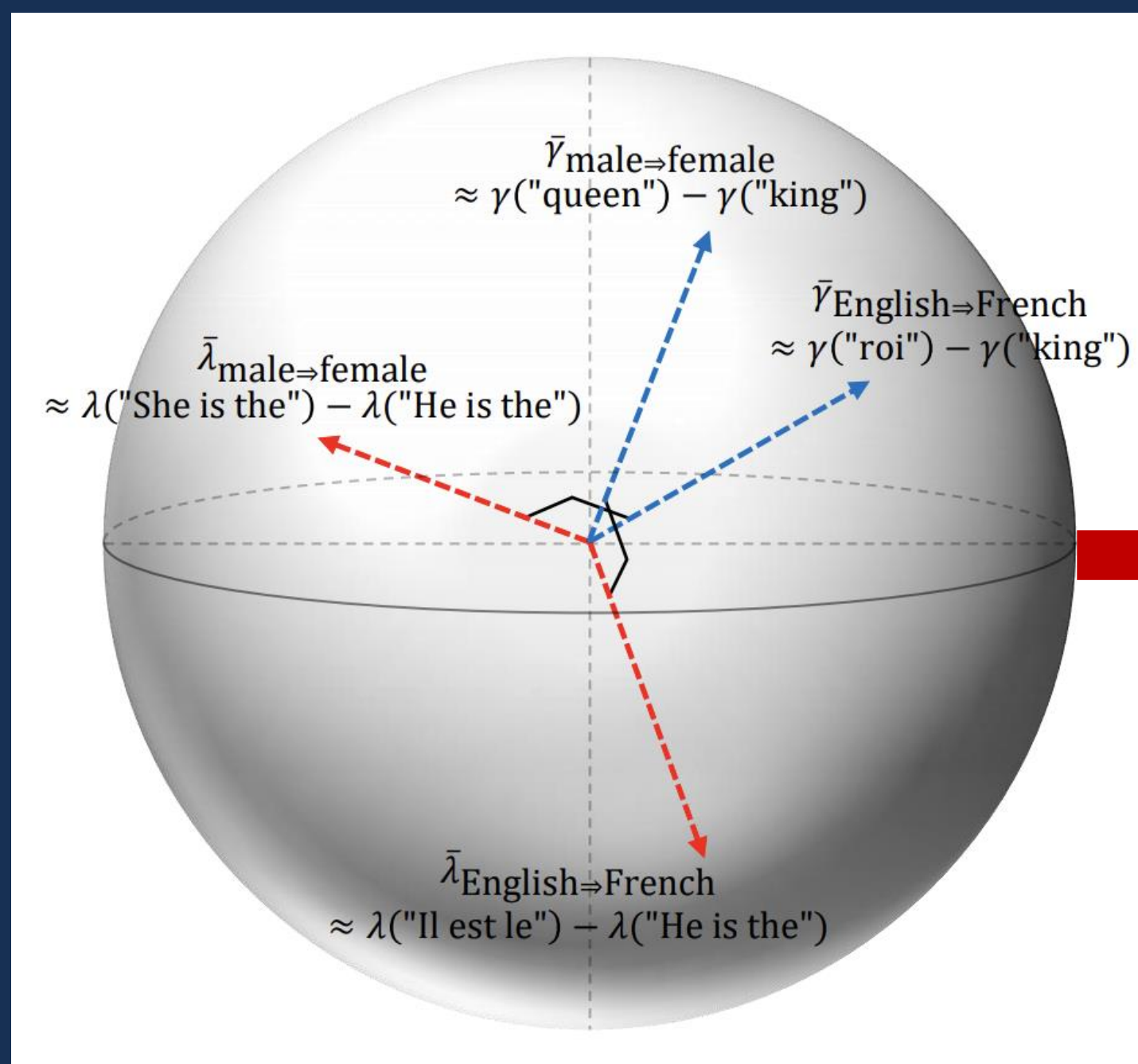




The transformation of concepts between contexts is not uniform but semantically meaningful and shared across models

Language Models Represent and Transform Concepts with Shared Geometry

Zhimin Hu, Lanhao Niu, Sashank Varma

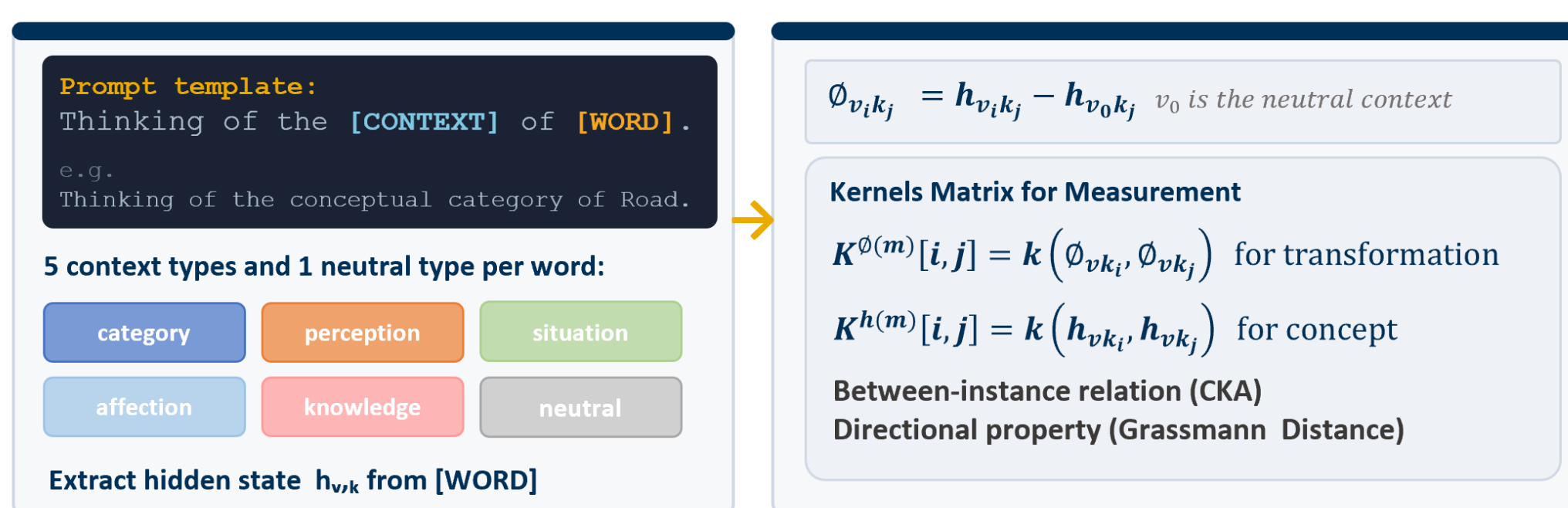
1 TL;DR

- Context moves each concept differently, not uniformly - in its own direction and magnitude
- This variation is semantically organized, not noise. They vary with lexical concreteness and density
- Models share a common geometry - not only in how concepts are represented, but in how context transforms them - across six model families

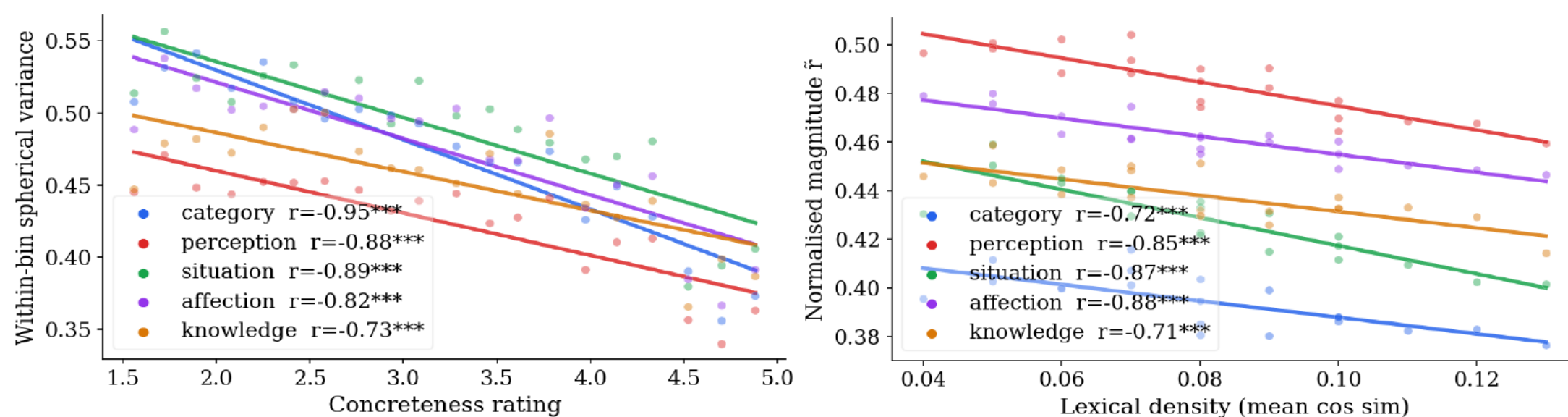
2 Background / Motivating Question

- Existing view treats a concept as a static vector
- Yet concepts appear in context, and context moves them.
- Does the way context transforms a concept have structure, or even shared across models?

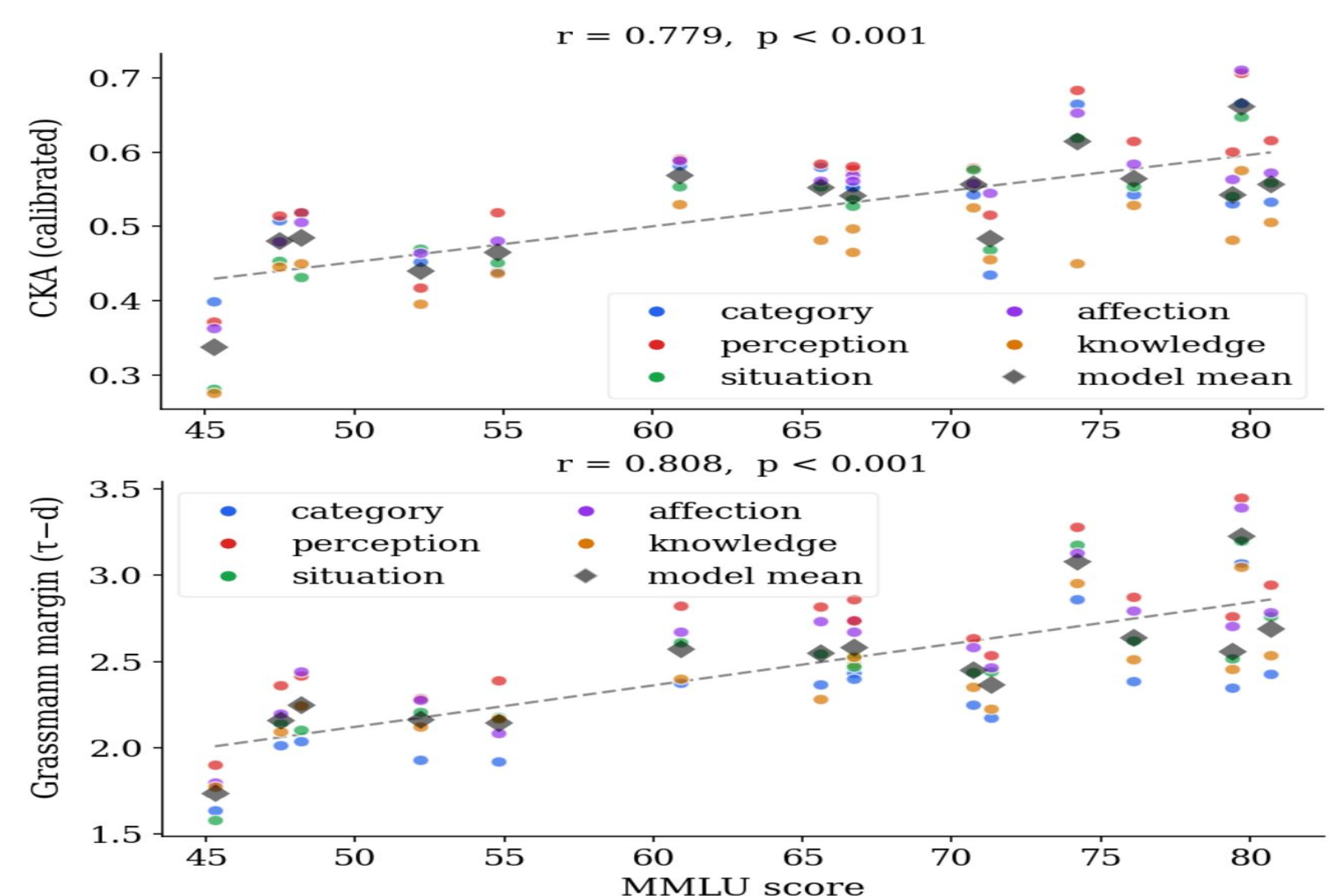
3 Measurement



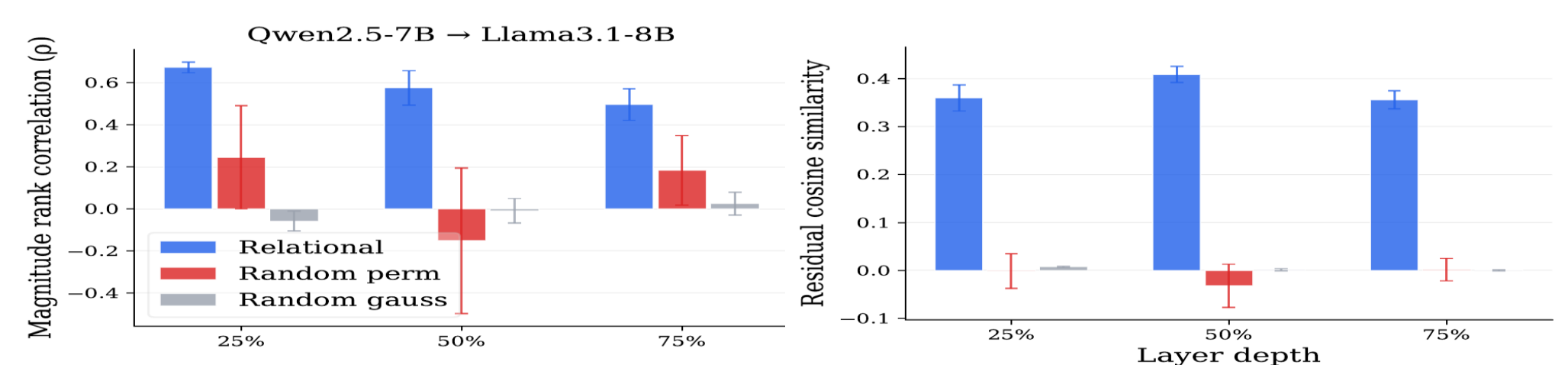
4 Share Geometry of Transformation



- More concrete concepts are displaced in more consistent directions across contexts.
- Concepts in denser neighborhoods (from Word2Vec) are displaced by smaller magnitudes.
- Shared displacement structure correlates more strongly with MMLU score than with parameter count.

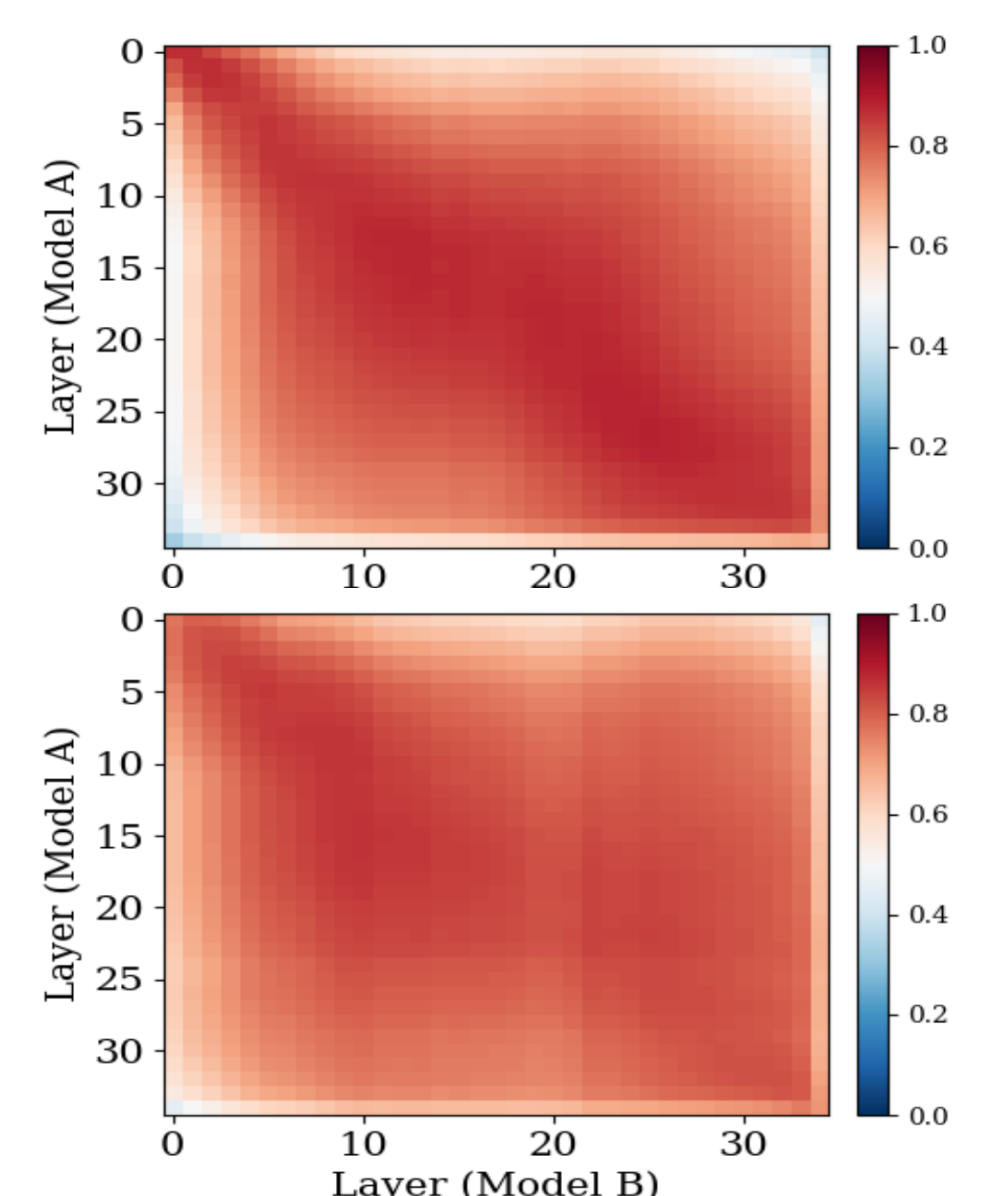


- Displacement structure from Model A predicts held-out displacements in Model B



4 Share Geometry of Concept

- The structure of various concepts under the same context (top) and the same concept under various contexts (down) is aligned (averaged across models)
- Alignments can not be reduced to surface statistical co-occurrence and scramble context degrade the alignments



5 Limitations and Discussion

- Scope of contexts and the size of vocabulary
- Steering and alignment methods manipulate something richer than a single direction—when we steer a representation, what exactly are we moving?
- Geometric structure support concepts to be simultaneously flexible and stable.



PAPER



CODE